Owen Liu

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INTERESTS

AI agents and agent evaluation, synthetic data for LLM training

EDUCATION

Stanford University Jun 2025 – Present
M.S. in Computer Science with Clinical Informatics Management (expected Summer 2027)

Columbia University, Research Assistant

Baruch College, City University of New York
B.S. in Mathematics, top 1%

CONFERENCES

Attendee, Conference on Language Modeling (**COLM 2026**), San Francisco Oct 2026

RESEARCH

Working papers

[1] Owen Yajie Liu, Kara Liu, Michael S. Bernstein. Evaluating numerical preprocessing methods for tabular representation learning. Working paper, 2026.

[2] Owen Yajie Liu. Trick or Treat: Can a fast AI tell which agent decisions are tricks? Working paper, 2026.

Open research projects I contributed to (not a listed author)

[3] Etash Guha, Ryan Marten, Sedrick Keh, et al. OpenThoughts: Data recipes for reasoning models. *arXiv preprint arXiv:2506.04178*, 2025. [link]

EXPERIENCE

Stanford CS329Z: Building AI Agents, Course Research Project Oct 2026 – Present
Trick or Treat: Can a fast AI tell which agent decisions are tricks? [2]

- **Execution-grounded Eval**: Label real SWE-agent decision points by **execution replay**; test whether fast models can label them at a glance with calibrated confidence.
- **Hard Negatives**: Perturbed "sneaky trick" twins (swapped flag, deleted observation, injected distractor), verified by replay.
- **Metrics**: Coverage at 95% precision, Brier/ECE and replay cost saved.

Stanford University, Growth Lead, AI Agents for Synthetic Data Jan 2025 – Dec 2025

- **Agent Benchmarks & Reasoning Data**: Contributor to **Terminal-Bench** and **OpenThoughts** (5M+ downloads), open projects for evaluating AI agents and training reasoning models [3].
- **Evaluation Matrix**: Prototyped with engineering and design an evaluation matrix judging research against product criteria; the demo shaped the roadmap.
- **Agent Prototyping**: Built and demoed LLM-agent and marketplace prototypes under time pressure; **3x hackathon winner at AGI House**.
- **Incubation**: Admitted to Stanford **StartX** incubator, Fall 2025 cohort (13 of 200 selected).

TATA Technology Inc., Co-Founder Sep 2020 – Dec 2024, Hong Kong

- **Two-sided Healthcare Marketplace**: Built a marketplace matching families of autistic children with vetted, trained providers. Grew from 0 to **20,000+ families** and **200+ providers** and community partners; company **acquired**.
- **Data Pipeline**: Built the pipeline that collected, cleaned and structured provider, family and intake-form data, the data behind every family–provider match.
- **Data Quality Gates**: Ran provider screening and onboarding by hand first, then codified it into repeatable workflows with explicit quality checks that scaled without the founder.
- **Go-to-market**: Owned partner-facing go-to-market end to end (positioning, onboarding funnels, activation), iterating weekly on conversion and retention.
- **Team**: Hired and led a **25-person** cross-functional team (product, engineering, operations, partnerships) on a $500K+ annual budget.

SELECTED AWARDS

3x Hackathon Winner, AGI House 2025
Stanford StartX Incubator, Fall 2025 cohort (13 of 200 selected) 2025
B.S. Mathematics, top 1%, Baruch College (CUNY)

LEADERSHIP

President, Board of Directors, Columbia University Alumni Sports League Jan 2020 – Jan 2022

SKILLS & OTHER

AI / LLM: LLM agents, agent evaluation, synthetic-data pipelines, SFT, prompt engineering
Languages: English (fluent), Chinese (native)
Sports: Former pro table tennis player and coach of nationally qualified youth; 4.5 DUPR pickleball